11/9/26 Solution:

Matrix-Chain:

$$m[i,j] = m[i,k] + m[k+1, j] + p_{i-1} p_k p_j$$

Choose k which gives min $m[i,j]$

$A_i \leftarrow p_{i-1} \times p_i$

$A_1 - 10 \times 100$

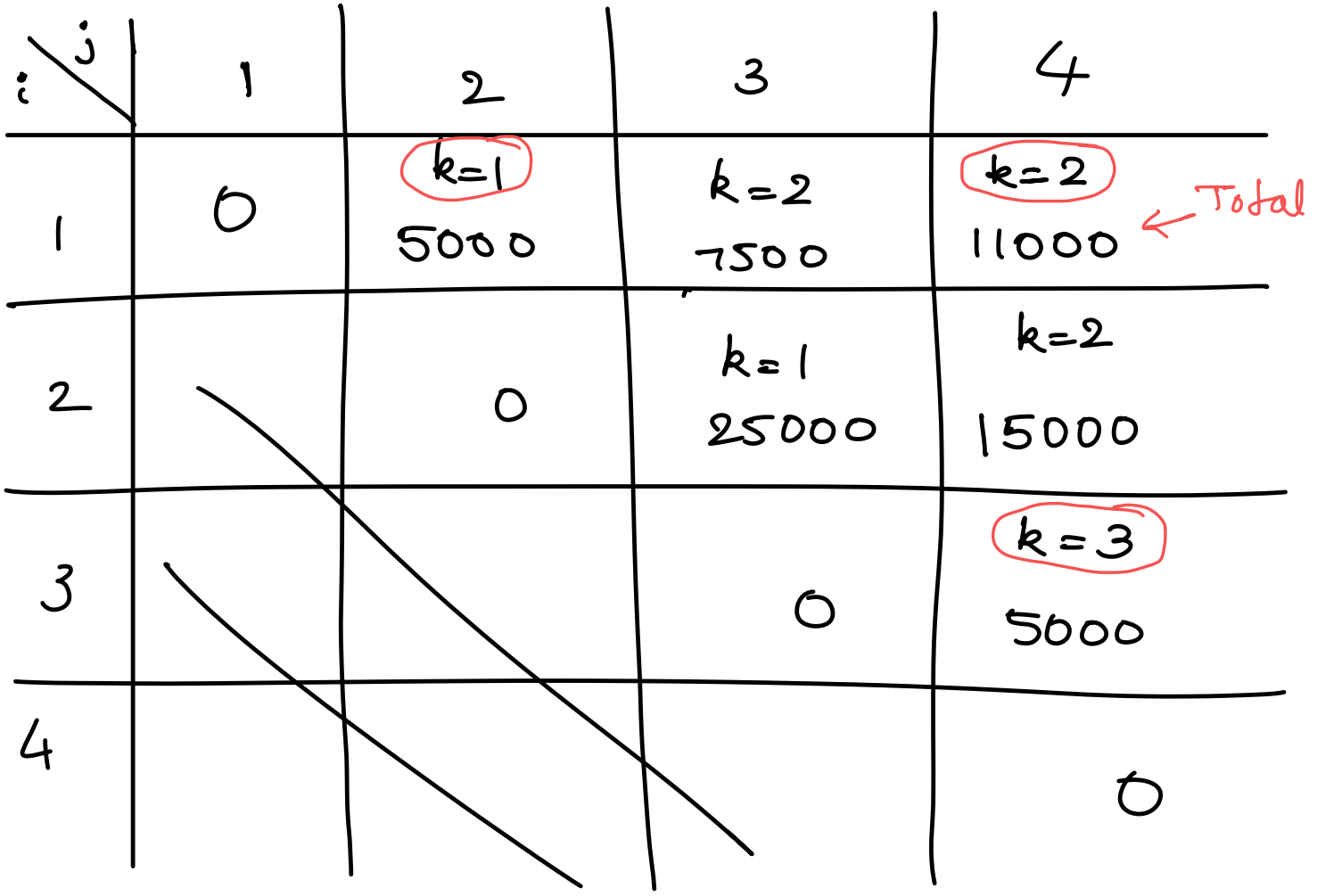
$A_2 - 100 \times 5$

$A_3 - 5 \times 50$

$A_4 - 50 \times 20$

"Parenthesize $A_1 A_2 A_3 A_4$".

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i
j
1
2
3
4
1
0
k=1
5000
k=2
7500
k=2
11000
Total.
2
0
k=1
25000
k=2
15000
3
0
k=3
5000
4
0

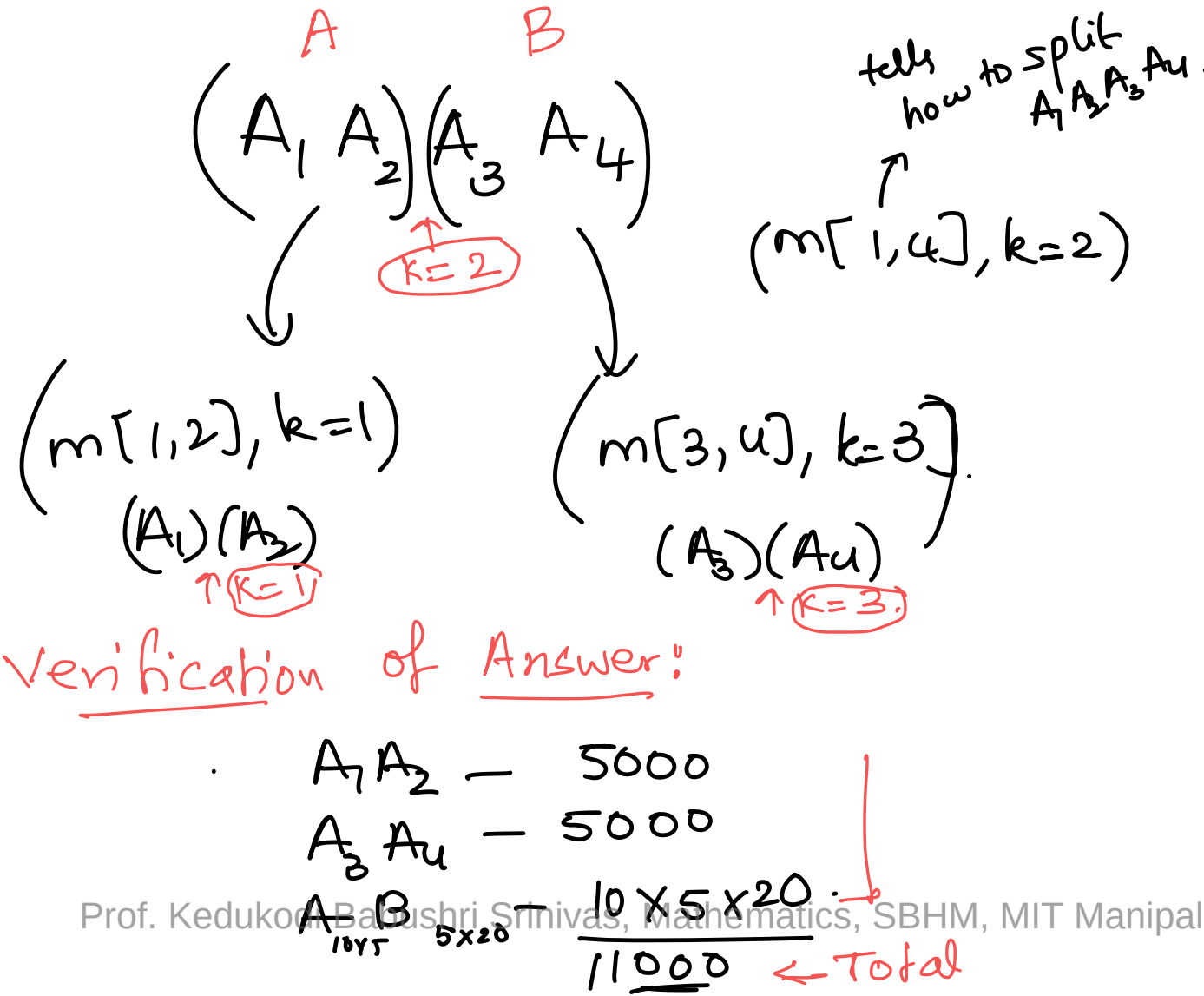
$$m[3,4] = m[3,3] + m[4,4] + p_2 p_3 p_4$$

$k=3$ $= 0 + 0 + 5 \times 50 \times 20$

$= 5000$

$$m[2,4] = \begin{cases} m[2,2] + m[3,4] + p_1 p_2 p_4 & k=2 \\ 0 + 5000 + 100 \times 5 \times 20 \\ = 15000 \quad (10000) \\ m[2,3] + m[4,4] + p_1 p_3 p_4 & k=3 \\ 25000 + 0 + 100 \times 50 \times 20 \\ = 125000 \quad (100 \times 1000 = 100000) \end{cases}$$

$$m[1,4] = \begin{cases} m[1,1] + m[2,4] + p_0 p_1 p_4 & k=1 \\ 0 \quad 15000 \quad 10 \times 100 \times 20 \\ = 35000 \quad 1000 \times 20 = 20000 \\ m[1,2] + m[3,4] + p_0 p_2 p_4 & k=2 \\ 5000 \quad 5000 \quad 10 \times 5 \times 20 \\ 10000 \quad 10 \times 100 \\ = 11000 \quad = 1000 \\ m[1,3] + m[4,4] + p_0 p_3 p_4 & k=3 \\ 7500 \quad 0 \quad 10 \times 50 \times 20 \\ 10 \times 1000 \\ = 17500 \quad 10000 \end{cases}$$



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Final Parenthesis:

